



FIRST LAW OF THERMODYNAMICS



Thermal Energy and Work:

Internal Energy and Heat (Q): In thermodynamics, thermal energy is associated with the internal energy of a system. Heat (Q) is the transfer of thermal energy between systems due to a temperature difference.

Heat Flow:

It flows spontaneously from a region of higher temperature to one of lower temperature. The rate at which heat is transferred can be quantified using the

Equation:

$$Q = mc\Delta T$$

Where Q is the heat transferred, m is the mass of the substance, c is the specific heat capacity of the substance, and ΔT is the temperature change.

Internal Energy:

"The sum of the random distribution of kinetic and potential energies within a system of molecules."

- The symbol for internal energy is U, and its SI unit is Joule (J).

Internal Energy & Temperature:

The internal energy of an object is intrinsically related to its temperature.

When a gas in a container is heated, the gas molecules move faster, increasing their kinetic energy.

In a solid, where molecules are tightly packed, heating causes the molecules to vibrate more.

In both liquids and solids, molecules have both kinetic and potential energy due to intermolecular forces that keep them close together.

However, ideal gas molecules are assumed to have no intermolecular forces, meaning they only possess kinetic energy and no potential energy. The (change in) internal energy of an ideal gas is equal to:

$$\Delta U = \left(\frac{3}{2}\right) k\Delta T$$

Therefore, the change in internal energy is proportional to the change in temperature:

$$\Delta U \propto \Delta T$$

Where:

ΔU = change in internal energy (J)

ΔT = change in temperature (K)

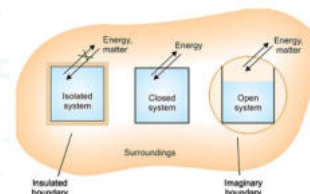
"As the container is heated up, the gas molecules move faster with higher kinetic energy and therefore higher internal energy."

Thermodynamic terms and basic concepts:

The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation is called thermodynamics.

Boundary:

The boundary is the physical or imaginary separation between the system and its surroundings.



Surroundings:

The surroundings refer to everything outside the boundaries of the system.

Thermodynamic Systems:

1. Open System:

In an open system, both heat and matter can be exchanged with the surroundings. For instance, steam (matter) can escape from the coffee, and heat can be lost to the surrounding air.

Example: A cup of hot coffee.

2. Closed System:

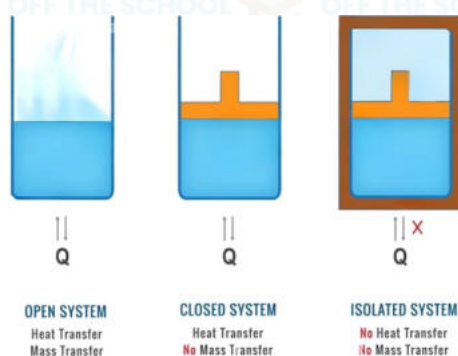
In a closed system, only heat can be exchanged with the surroundings, but matter cannot. In this example, heat can still transfer through the pot, but the water vapor (matter) is trapped inside the pot and cannot escape.

Example: A pot with a lid.

3. Isolated System:

In an isolated system, neither heat nor matter can be exchanged with the surroundings. A thermos flask is designed to prevent heat transfer, keeping the contents inside hot or cold for a long period without any exchange with the Environment.

Example: A thermos flask.



Work Done by a Gas:

When a gas expands, it does work on its surroundings by exerting pressure on the walls of the container it's in. This is important, for example, in a steam engine where expanding steam pushes a piston to turn the engine. The work done when a volume of gas changes at constant pressure is defined as:

$$W = P\Delta V$$

Where:

W = work done (J)

P = external pressure (Pa)

V = volume of gas (m³)

For a gas inside a cylinder enclosed by a moveable piston, the force exerted by the gas pushes the piston outwards, therefore, the gas does work on the piston.

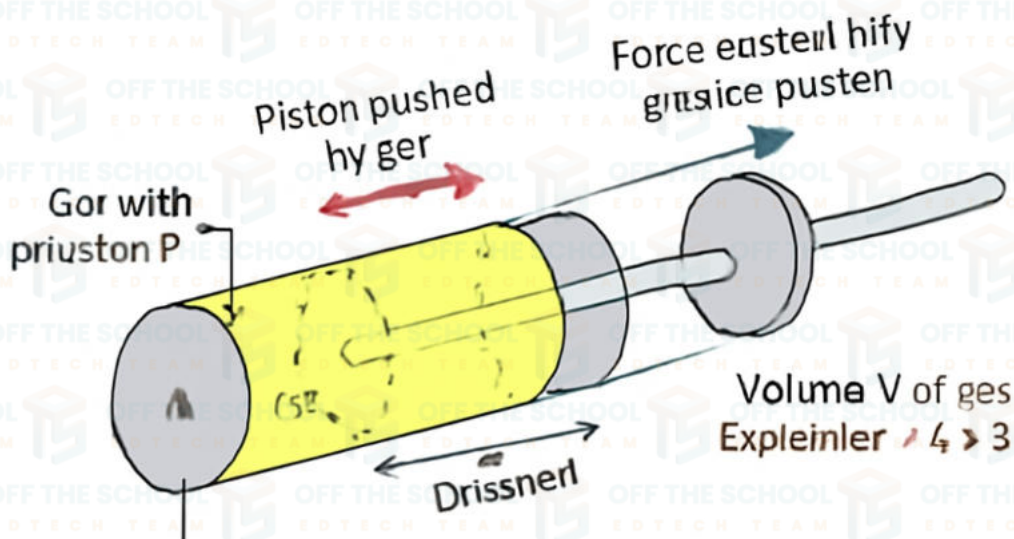
Gas with pressure P

Volume V of gas expanded = $A \times S$

Cross sectional area

The gas expansion pushes the piston a distance S

Figure 16.5 Work Done by a Gas



The gas expansion pushes the piston a distance S

Derivation:

The volume of gas is at constant pressure. This means the force F exerted by the gas on the piston is equal to:

$$F = P \times A$$

The definition of work done is:

$$W = F \times d$$

The displacement of the gas d multiplied by the cross-sectional area A is the increase in volume ΔV of the gas:

$$W = P \times A \times d$$

Since, $A \times d = \Delta V$

$$W = P \Delta V$$

Where: ΔV = increase in the volume of the gas in the piston when expanding (m^3)

First Law of Thermodynamics, Internal Energy, and the Heating of the System & Conservation of Energy

First law of thermodynamics:

The first law of thermodynamics states that,

"The change in internal energy of a system equals the net heat transfer into the system minus the net work done by system"

In equation form,

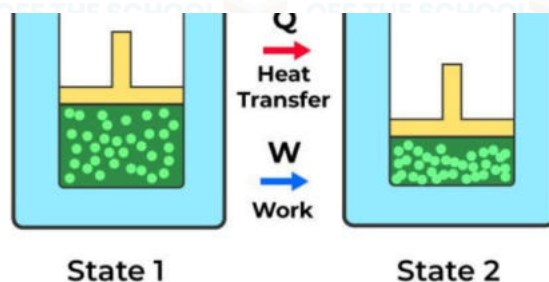
$$\Delta U = \Delta Q - W$$

Where:

$\Delta U \rightarrow$ change in internal energy of the system

$\Delta Q \rightarrow$ net heat transferred into the system

$W \rightarrow$ net work done by the system



Applications of the first law thermodynamics:

Applications of first law of thermodynamics are all based on the principle that

"The three forms of energy: internal energy; heat; and work can be inter converted."

- ✓ The processes in which any one of the three terms of the equation is zero.
- ✓ The processes in which any one of the state variables P, V, T of the system are held constant.

(a) Isochoric Process: Constant volume

"The thermodynamics process during which the volume of the system remains constant is called isochoric process".

We consider the gas contained in a cylinder having a conducting base and non-conducting walls and with a fixed piston at the end as shown in the figure. Let heat ΔQ be imparted to the gas. The gas is then heated at constant volume. The pressure of the gas increases from P_1 to P_2 while its temperature increases from T_1 to T_2 .

Since the system neither expands nor contracts, work is neither done by the system nor on the system i.e $\Delta W = 0$.

Using the first law of thermodynamics equation:

We have:

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U$$

This means that in an isochoric process the entire amount of heat supplied to the gas is converted into the internal energy of the gas.

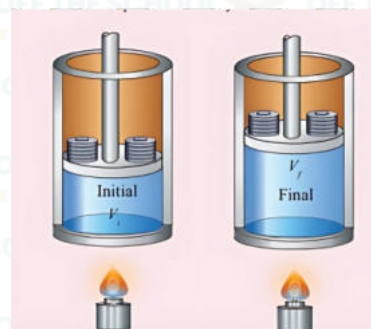
(b) Isobaric Process:

"The thermodynamics process during which the pressure is kept constant is called an isobaric process"

Isobaric expansion of a system is often used to convert heat into work. Practically all heat engines depend on the transformation of heat into work. Consider a gas contained in a cylinder having a conducting base and non-conducting walls and frictionless piston of cross-sectional area (A)

When the gas is heated, a certain amount of Heat energy is transferred into the system.

The gas expands and moves the piston outward.



Temperature changes from T_1 to T_2 and Volume changes from V_1 to V_2

If the displacement of the piston is kept very small, the pressure of the gas will not change much and can be considered constant.

The work done by the gas which expands at constant pressure is:

$$\Delta W = P(V_2 - V_1)$$

$$\Delta W = P\Delta V$$

Using the first law of thermodynamics equation:

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U + P\Delta V$$

The work performed by the expanding or contracting gas comes from one or both sources:

- > Heat supplied to the gas and
- > the internal energy of the gas

The graph of isobaric process is called an “isobar”, which is a straight line, parallel to the volume axis.

(c) Isothermal Process:

“The thermodynamics process which is carried out in such a way that a system undergoes changes but its temperature remains constant is called an isothermal process”.

During isothermal expansion some work is done by the gas in pushing up the piston in the cylinder. Since the temperature remains constant, there is no change in the internal energy of the gas, that is,

$$\Delta U = 0$$

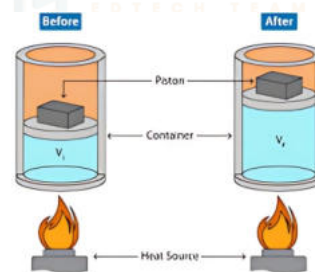
$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = 0 + \Delta W$$

$$\Delta Q = \Delta W$$

This shows that:

If the gas expands and does external work, an equal amount of heat has to be supplied in order to maintain its temperature constant.

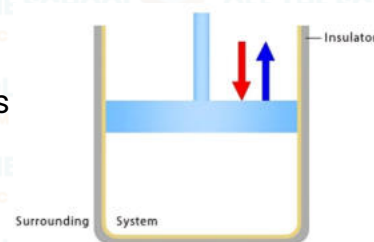


(d) Adiabatic Process:

"The thermodynamics process during which no heat enters or leaves a system is called an adiabatic process".

For every adiabatic process $\Delta Q = 0$, A truly adiabatic process is an ideal one which cannot be realized. However, the

flow of heat may be prevented either by surrounding the system with a thick layer of heating insulating material such as cork, asbestos, or by performing the process very quickly. The flow of heat requires finite time, so any process performed quickly enough will be practically adiabatic. To perform an adiabatic process on a gas, we consider the gas to be contained in a completely insulated cylinder with a movable piston at the end.



Heat capacity:

Amount of heat required to raise the temperature of a substance through 1°C or 1K is known as heat capacity.

$$\Delta Q = c \Delta T$$

Where "c" is the Heat capacity.

Specific Heat:

Amount of heat required to raise the temperature of unit mass of a substance through 1°C or 1K is known as specific heat capacity.

$$\Delta Q = m c \Delta T$$

Where "c" is the Specific heat capacity.

Molar Specific Heat:

Amount of heat required to raise the temperature of unit mole of a substance through 1°C or 1K is known as molar specific heat capacity.

Relationship between C_p and C_v for an Ideal Gas:

Relation between Specific Heat at Constant Pressure and Specific Heat at Constant Volume:

Consider a gas in a container if we heat the gas so that gas is allowed to expand and pressure of the gas remain constant all the heat energy will be used to increase the internal energy of the gas and to do work than according to first law of thermodynamics.

$$\Delta Q = \Delta U + \Delta W$$

Since:

Above equation will become:

$$\Delta Q_p = \Delta U + P\Delta V \text{ ---- (1)}$$

Using general gas law:

$$PV = nRT$$

Hence:

$$P\Delta V = nR\Delta T$$

Putting in Equation (1) we get:

$$\Delta Q_p = \Delta U + nR\Delta T$$

Since:

$$\Delta Q_p = nC_p\Delta T$$

Above equation will become:

$$nC_p\Delta T = \Delta U + nR\Delta T \text{ -- (a)}$$

Consider a gas in a container if we heat the gas so that gas is not allowed to expand so that volume of the gas remain constant and all the heat energy will be used to increase the internal energy of the gas than according to first law of thermodynamics.

expand so that volume of the gas remain constant and all the heat energy will be

$$\Delta Q = \Delta U + \Delta W$$

Since work done is Zero:

$$\Delta Q_v = \Delta U$$

Since:

$$\Delta Q_v = nC_v\Delta T$$

Above equation will become:

$$\Delta Q_v = nC_v\Delta T$$

$$\Delta U = nC_v\Delta T \text{ --b}$$

Putting equation (a) in equation (b) we get:

$$nC_p\Delta T = nC_v\Delta T + nR\Delta T$$

Or

$$nC_p\Delta T = n\Delta T(C_v + R)$$